

1. Administrative & Study Identification

Full Study Title: Open-label, Long-term Safety Study of Secukinumab in Polymyalgia Rheumatica (PMR) **Short Title/Acronym:** REPLENISH-EXT **Protocol Number:** CAIN457C22301E1 **NCT Number:** NCT06331312 **Sponsor Name:** Novartis Pharmaceuticals / Novartis Pharma AG **Investigational Product:** Secukinumab (AIN457) **Phase of Development:** Phase III **Medical Monitor:** Novartis Medical Monitor (Clinical Operations)

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To assess the long-term safety and tolerability of secukinumab when administered long-term in patients with polymyalgia rheumatica (PMR). **Secondary Objectives:** To evaluate the long-term efficacy of secukinumab among patients with PMR, with a particular focus on achieving sustained glucocorticoid reduction without triggering disease flares.

2.2 Background & Rationale To address the need for safe, long-term disease control and sustained glucocorticoid reduction in patients with PMR. The study evaluates whether patients who have completed the 52-week Treatment Period in the core secukinumab study (Study CAIN457C22301) and experienced a relapse during the treatment-free follow-up period can safely derive long-term benefit from continued IL-17A inhibition therapy without relying on high-dose rescue treatments.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Interventional (Extension Study) **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration **Trial Status:** Recruiting **Target N:** 300 participants (Ages 50-100) **Number of Sites:** Multicenter (Approx. 115 locations globally) **Estimated Study Start:** 28-Jun-2024 **Estimated Primary Completion:** 22-Aug-2028

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Participants aged 50 to 100 years who have completed the 52-week Treatment Period as per protocol in a Novartis core study of secukinumab in PMR patients (Study CAIN457C22301). 2. Participants who have experienced a relapse

during the treatment-free follow-up period of the core study and who have not been on rescue treatment. 3. The participant would potentially derive benefit from secukinumab, and the benefit outweighs the risk, based on the investigator's judgement.

4.2 Exclusion Criteria 1. Use of prohibited medications, as specified in the protocol. 2. History of ongoing, chronic, or recurrent infectious disease (i.e., human immunodeficiency virus (HIV), hepatitis B (HBV), hepatitis C (HCV), active tuberculosis infection (TB)). 3. History of lymphoproliferative disease or any known malignancy or history of malignancy of any organ system within the past 5 years (except for basal cell carcinoma or actinic keratosis that have been treated with no evidence of recurrence in the past 3 months, carcinoma in situ of the cervix, or non-invasive malignant colon polyps that have been removed). 4. Administration of live vaccinations (e.g., monkey pox vaccine, oral polio vaccine, varicella/zoster vaccines) within 6 weeks prior to Baseline. 5. Subjects whose participation in the extension study could expose them to an undue safety risk.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Secukinumab 300 mg (administered as 2 x 150mg/1mL pre-filled syringes). Administered every 4 weeks during Treatment Period 1 (TP1) (Weeks 16 and 20 visits are optional on-site visits and needed when participants are unwilling/uncomfortable to self-administer study treatment at home/offsite). Followed by administration/resupply every 12 weeks during Treatment Period 2 (TP2), though participants unwilling/uncomfortable to self-administer study treatment can continue to visit the site every 4 weeks for drug administration. **Route of Administration:** Subcutaneous (SC) injection. **Duration of Treatment:** Up to 2 years total Treatment Period (TP1: Baseline to Week 24; TP2: Post-Week 24 up to 2 years), followed by a 16-week treatment-free follow-up period.

5.2 Concomitant & Prohibited Therapy Permitted: Standard ongoing stable background therapies that do not conflict with the protocol. **Prohibited:** Use of protocol-specified prohibited medications; live vaccinations during the study.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Up to -28 to 0	Informed Consent, Medical History, Eligibility Confirmation (from core study)
Baseline	Day 0	First Dose Administration of secukinumab, Baseline Safety Labs
Interim 1 (TP1 & TP2)	Week 4 to Year 2	TP1: Visits every 4 weeks until Week 24 (Weeks 16 and 20 are optional on-site if uncomfortable with at-home self-administration). TP2: Visits every 12 weeks for drug resupply, or

every 4 weeks if unwilling to self-administer. Continuous safety tracking. | | **End of Study (EoS)** | Week 120 (approx) | 20 weeks post last dose of secukinumab (16-week treatment-free follow-up): Final safety evaluations, Exit Interview |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Incidences of treatment emergent adverse events (AEs) and serious adverse events (SAEs). The number and percentage of participants with treatment-emergent AEs/SAEs will be summarized. **Measurement Method:** Clinical assessment, physical examinations, and continuous AE/SAE reporting logs. No hypothesis testing will be performed.

7.2 Secondary & Exploratory Endpoints 1. Long-term efficacy and disease control of secukinumab in PMR. 2. Sustained glucocorticoid reduction without triggering PMR disease flares.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Method of collection includes patient reporting, physical examinations, and laboratory evaluations during scheduled onsite visits and remote check-ins. **Serious Adverse Events (SAEs):** Reported continuously per standard pharmacovigilance safety guidelines (typically within 24 hours of site awareness). **Data Monitoring Committee (DMC):** Internal/External safety monitoring boards governed by Novartis to review long-term safety signals.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all participants who received at least one dose of the investigational product). **Sample Size Justification:** $N=300$; as a long-term safety extension, the sample size is driven by the eligible population rolling over from the core study rather than powered hypothesis testing. **Handling of Missing Data:** Standard descriptive statistics with data handling as defined in the overarching protocol; no formal imputation for hypothesis testing for the primary safety outcome.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite and offsite monitoring visits for drug accountability and safety profile tracking. **Audit Trail:** Standard tracking of eCRF locking, source data verification, and secure tracking of SC injection compliance.

11. Study Identifiers & Governance

Primary ID: CAIN457C22301E1 **Registry Number:** NCT06331312 **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** REPLENISH-EXT **Official Regulatory Title:** A Multi-center, Open-label Extension Study of Subcutaneous Secukinumab to Evaluate the Long-term Safety and Tolerability in Polymyalgia Rheumatica (PMR)

12. Clinical Framework (PMR Specific)

Primary Condition: Polymyalgia Rheumatica (PMR) **Diagnosis Threshold:** Relapse during the treatment-free follow-up period of the CAIN457C22301 core study. **Medical Methodology:** Interleukin-17A (IL-17A) inhibition therapy (Secukinumab) **Optimization Target:** Sustained glucocorticoid reduction and prevention of disease flares.

13. Study Procedures & Methodology

Management Pathway: Long-term safety extension and biologic maintenance. **Education Protocol (HCP):** Protocol training on long-term AE identification and SC medication dispensation. **Education Protocol (Patient):** Self-administration training for SC pre-filled syringes (for at-home use where applicable) and safety reporting instructions. **Quality Audit Mechanism:** Routine drug accountability checks and protocol adherence tracking via eCRF.

14. Observation & Follow-Up Schedule

Total Duration: Up to 120 weeks (Up to 2 years treatment + 16-week treatment-free follow-up) **Onsite Visit Cadence:** Every 4 weeks during TP1 (up to Week 24, with Weeks 16 and 20 being optional on-site); every 12 weeks during TP2 (or every 4 weeks if uncomfortable with self-administration). **Remote Monitoring Frequency:** Intermittent remote monitoring aligned with at-home SC administrations. **Checklist Review Interval:** Safety tracking at every scheduled dispensation visit (Q4W or Q12W).

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]				Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]				